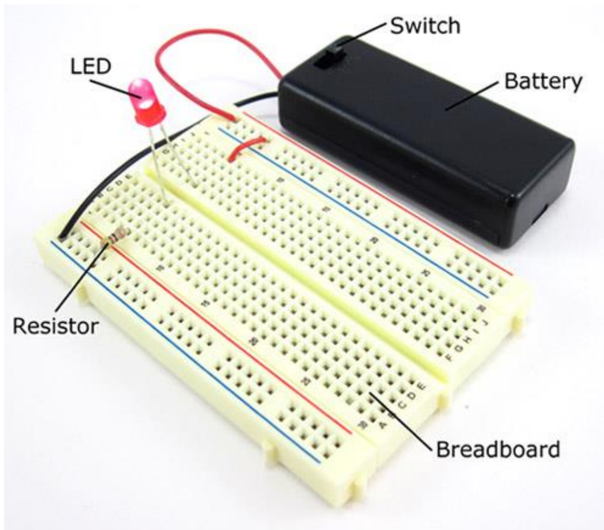
Introduction to Breadboard

This session will give you a basic introduction to breadboard and explain how to use in Lab experiments. This video link can help you learn more about breadboard: https://youtu.be/6WReFkfrUIk

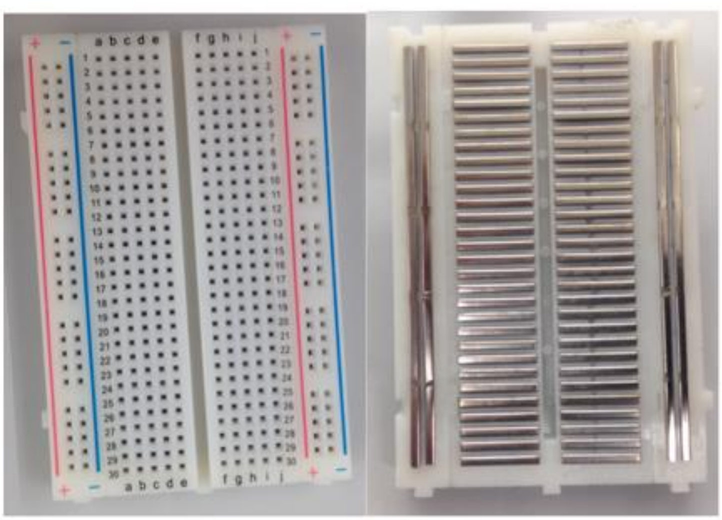
What is a breadboard?

A breadboard is a rectangular plastic board with a bunch of tiny holes in it. These holes let you easily insert electronic components like this one with a battery, switch, resistor, and an LED (light-emitting diode).

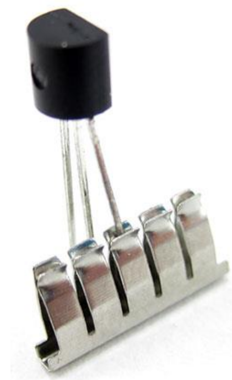


The connections are not permanent, so it is easy to *remove* a component if you make a mistake, or just start over and do a new project. This makes breadboards great for beginners who are new to electronics.

What is inside a breadboard?



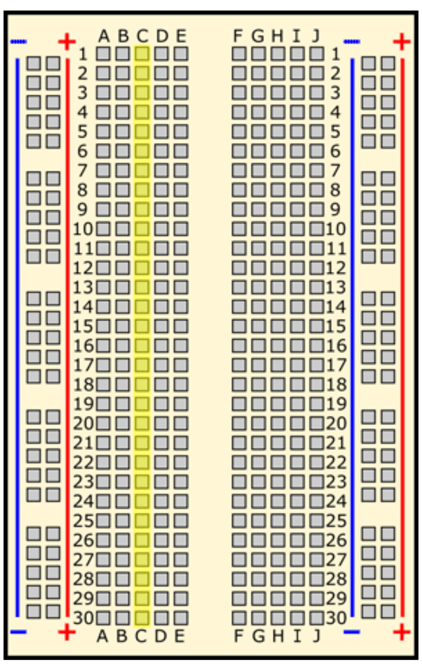
The leads can fit into the breadboard because the *inside* of a breadboard is made up of rows of tiny metal clips. This is what the clips look like when they are removed from a breadboard. When you press a component's lead into a breadboard hole, one of these clips grabs onto it.



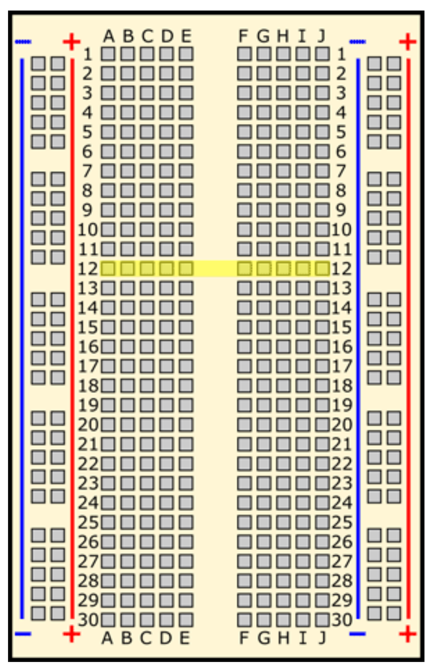
Breadboard labels: rows, columns, and buses

What do the letters and numbers on a breadboard mean?

Most breadboards have some numbers, letters, and plus and minus signs written on them. What does all that mean? These labels help you locate certain holes on the breadboard so you can follow directions when building a circuit. For example, all of the highlighted holes are in "column C."



All of the highlighted holes are in "row 12."



"Hole C12" is where column C intersects row 12.

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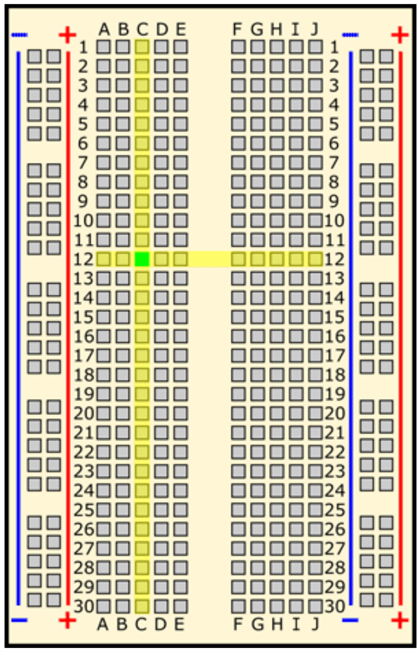
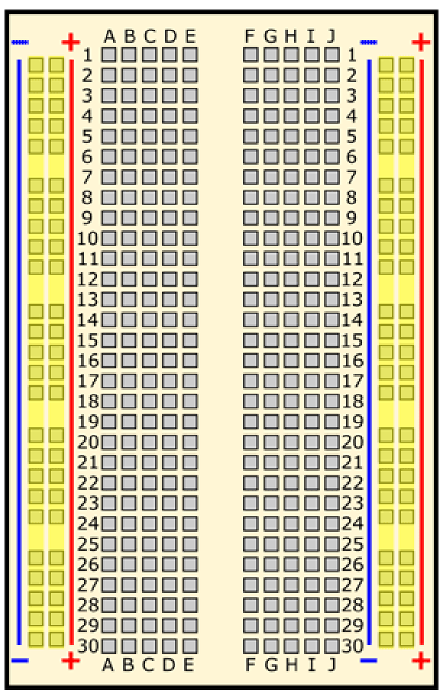


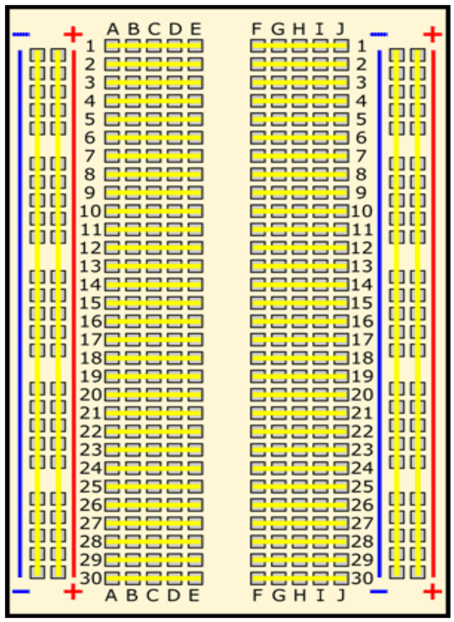
What do the colored (blue and red) lines and plus and minus signs mean? What about the long strips on the side of the breadboard, highlighted in yellow here?



These strips are typically marked by red and blue (or red and black) lines, with plus (+) and minus (-) signs, respectively. They are called the **buses**, also referred to as **rails**, and are typically used to supply electrical power to your circuit when you connect them to a battery pack or other external power supply.

How are the holes connected?

The inside of a breadboard is made up of sets of five metal clips. This means that each set of five holes forming a half-row (columns A–E or columns F–J) is electrically connected. For example, hole A1 is electrically connected to holes B1, C1, D1, and E1. It is *not* connected to hole A2, because that hole is in a different row. It is also *not* connected to holes F1, G1, H1, I1, or J1, because they are on the other "half" of the breadboard—the clips are not connected across the gap in the middle. This image shows which holes are electrically connected in a typical half-sized breadboard, highlighted in yellow lines.

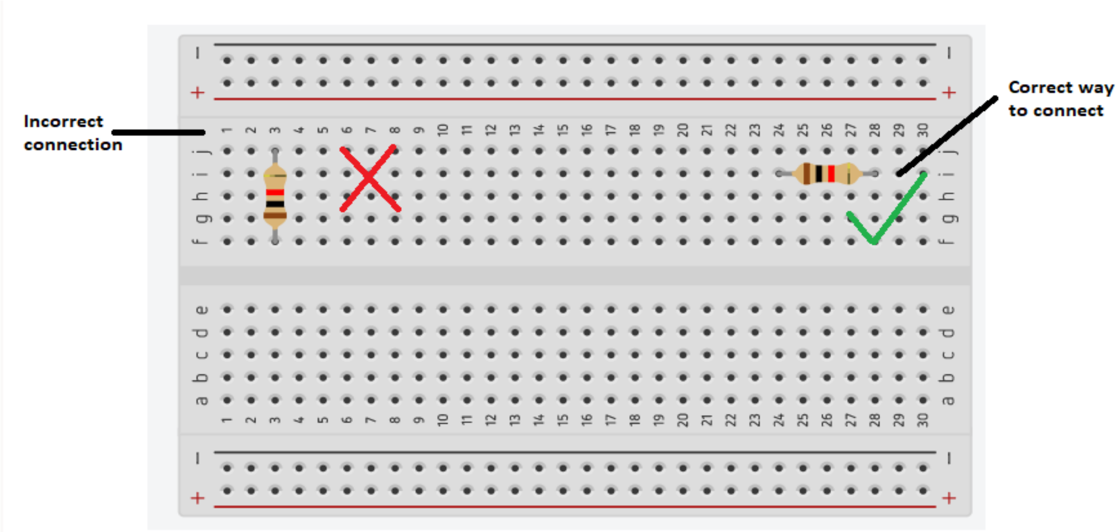


Common mistakes

Sometimes you may wonder why your circuit is not working! Most probably you made a mistake! Connection mistakes on breadboards are not vital, since connections are temporary. You can easily rearrange the circuit by removing or replacing the components. But you must pay attention to some common mistakes and avoid them to save time and finish your work smoothly. Here are some mistakes you need to be aware of:

Connecting leads of one component in the same row

Each component has at least two leads, and each lead has a different purpose in the circuit. Avoid putting the leads of one component in the same row to prevent a short circuit as mentioned in below figure. The resistor on the left side shows that its one leg is connected to its other leg (i.e. it is connected to itself and we called such a connection short connection)



Touching leads

Another reason for a short the circuit is when components touch each other accidentally. Some components have long leads, try to connect them apart from each other to avoid undesired lead contact.